Third-Body Perturbation in the Tidal Limit

Eric Pfahl — 07/20/01

1 Equations of Motion

Consider three particles of mass M_1, M_2, and M_3 with [instantaneous] positions $\boldsymbol{r}_1$, $\boldsymbol{r}_2$, and $\boldsymbol{r}_3$ in an inertial frame of reference (see Fig. 1). The equations of motion for particles 1 and 2, determined by their mutual gravitational attraction and their interaction with particle 3, are given by

$$\ddot{\boldsymbol{r}}_1 = -\frac{GM_2}{r^3}\boldsymbol{r} - \frac{GM_3}{s_1^3}\boldsymbol{s}_1 \; ; \tag{1}$$

$$\ddot{\boldsymbol{r}}_2 = +\frac{GM_1}{r^3}\boldsymbol{r} - \frac{GM_3}{s_2^3}\boldsymbol{s}_2 \; , \tag{2}$$

where $\boldsymbol{s}_i = \boldsymbol{r}_i - \boldsymbol{r}_3$. Subtracting eqs. (1) and (2), we find

$$\ddot{\boldsymbol{r}} + \frac{G(M_1+M_2)}{r^3}\boldsymbol{r} = -GM_3\left(\frac{\boldsymbol{s}_1}{s_1^3} - \frac{\boldsymbol{s}_2}{s_2^3}\right) \; , \tag{3}$$

where $\boldsymbol{r} = \boldsymbol{r}_1 - \boldsymbol{r}_2$. The HS of eq. (3), sometimes called the "disturbing function," constitutes the perturbative acceleration on the relative motion of particles 1 and 2. Clearly, in the limit that the RHS vanishes, the motion of particles 1 and 2 is purely Keplerian.

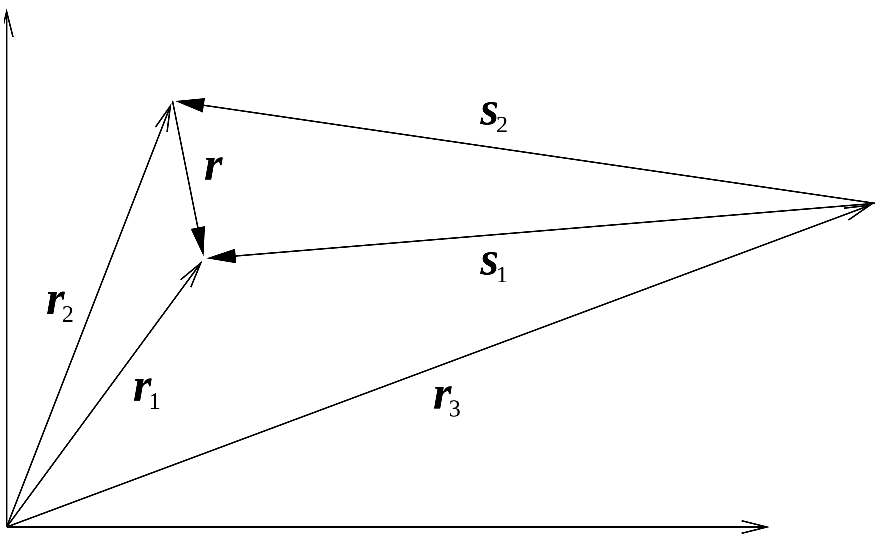

Figure 1: The geometry of the problem.

In the limit where $r \ll s_1, s_2$, the so-called "tidal limit," we can readily approximate the disturbing function. In the case where $M_2 \ll M_1, M_3$ it is natural to choose r/s_1 as the small expansion parameter. Writing $\boldsymbol{s}_2 = \boldsymbol{s}_1 - \boldsymbol{r}$, we have

$$\frac{\boldsymbol{s}_2}{s_2^3} \equiv \frac{\boldsymbol{s}_1 - \boldsymbol{r}}{|\boldsymbol{s}_1 - \boldsymbol{r}|^3} = \frac{\boldsymbol{s}_1 - \boldsymbol{r}}{s_1^3}\left[1 + \left(\frac{r}{s_1}\right)^2 - 2\left(\frac{r}{s_1}\right)u\right]^{-3/2} , \tag{4}$$

where $u \equiv \hat{\boldsymbol{r}} \cdot \hat{\boldsymbol{s}_1}$. If we expand the term in brackets to first order in (r/s_1), eq. (4) becomes

$$\frac{\boldsymbol{s}_2}{s_2^3} = \frac{\boldsymbol{s}_1 - \boldsymbol{r}}{s_1^3}\left[1 + 3\left(\frac{r}{s_1}\right)u + O\left(\frac{r}{s_1}\right)^2\right] . \tag{5}$$

Substituting eq. (5) into eq. (3), we obtain the equation of motion of two bodies in the tidal field of a distant third body:

$$\ddot{\boldsymbol{r}} + \frac{G(M_1 + M_2)}{r^3}\boldsymbol{r} = -\frac{GM_3}{s_1^2}\left(\frac{r}{s_1}\right)[\hat{\boldsymbol{r}} - 3\hat{\boldsymbol{s}}_1 u] , \tag{6}$$

where terms proportional to $(r/s_1)^2$ have been dropped.

2 Potential Formulation

The perturbative acceleration (RHS of eq. [3]) can be written as the gradient of a potential, V. It is easily verified that

$$\frac{\boldsymbol{s}_1}{s_1^3} - \frac{\boldsymbol{s}_2}{s_2^3} = \frac{\partial}{\partial \boldsymbol{r}}\left(\frac{\boldsymbol{r} \cdot \boldsymbol{s}_1}{s_1^3} - \frac{1}{s_2}\right) , \tag{7}$$

where $\partial/\partial\boldsymbol{r}$ denotes the gradient with respect to the relative coorinates $\{\hat{\boldsymbol{x}}_i \cdot \boldsymbol{r}\}$. Operationally, the gradient is computed by varying the position of particle 2, while holding particles 1 and 3 fixed. Therefore, the perturbative potential is given by

$$V = GM_3\left(\frac{\boldsymbol{r} \cdot \boldsymbol{s}_1}{s_1^3} - \frac{1}{s_2}\right) , \tag{8}$$

so that the perturbing acceleration is simply $-\partial V/\partial \boldsymbol{r}$.

Now, recall the generating function for the Legendre polynomials:

$$(1 + \epsilon^2 - 2\epsilon u)^{-1/2} = \sum_{l=0}^{\infty} \epsilon^l P_l(u) . \tag{9}$$

The first four Legendre polynomials are

$$\begin{aligned} P_0(u) &= 1 \\ P_1(u) &= u \\ P_2(u) &= \tfrac{1}{2}(3u^2 - 1) \\ P_3(u) &= \tfrac{1}{2}(5u^3 - 3u) . \end{aligned} \tag{10}$$

It is now a simple matter to expand $s_2^{-1} = |\boldsymbol{s}_1 - \boldsymbol{r}|^{-1}$ as a series in Legendre polynomials, for $r < s_1$:

$$|\boldsymbol{s}_1 - \boldsymbol{r}|^{-1} = \frac{1}{s_1}\left[1 + \left(\frac{r}{s_1}\right)^2 - 2\left(\frac{r}{s_1}\right)u\right]^{-1/2} = \frac{1}{s_1}\sum_{l=0}^{\infty}\left(\frac{r}{s_1}\right)^l P_l(u) \tag{11}$$

With this expansion in hand, V becomes

$$V = \frac{GM_3}{s_1}\left[\left(\frac{r}{s_1}\right)u - 1 - \left(\frac{r}{s_1}\right)u - \sum_{l=2}^{\infty}\left(\frac{r}{s_1}\right)^l P_l(u)\right] . \tag{12}$$

Note that the terms proportional to u^1 (dipole terms) cancel and we are left with

$$V = -\frac{GM_3}{s_1}\left[1 + \left(\frac{r}{s_1}\right)^2 \frac{1}{2}(3u^2 - 1) + \sum_{l=3}^{\infty}\left(\frac{r}{s_1}\right)^l P_l(u)\right] . \tag{13}$$

If we neglect all terms higher than quadrupole order ($l = 3$ and beyond) and compute the gradient, $-\partial V/\partial \boldsymbol{r}$, we recover the tidal acceleration on the RHS of eq. (6).